1.13 Roots of Polynomials: Using Substitutions

Worked Solutions

1 a)

$$3x^2 - 4x - 6 = 0$$

If $x = \alpha$ is a root of the original equation and we want an equation with the root $w = 2\alpha$, then we can rearrange to get $\alpha = \frac{1}{2}w$ and sub $\frac{1}{2}w$ into the original equation.

$$3\left(\frac{1}{2}w\right)^2 - 4\left(\frac{1}{2}w\right) - 6 = 0$$
$$\frac{3}{4}w^2 - 2w - 6 = 0$$
$$3w^2 - 8w - 24 = 0$$

1 b)

$$3x^2 - 4x - 6 = 0$$

If $x = \alpha$ is a root of the original equation and we want an equation with the root $w = 1 - \alpha$, then we can rearrange to get $\alpha = 1 - w$ and sub $1 - w$ into the original equation.

$$3(1-w)^2 - 4(1-w) - 6 = 0$$
$$3w^2 - 6w + 3 - 4 + 4w - 6 = 0$$
$$3w^2 - 2w - 7 = 0$$

1 c)

$$3x^2 - 4x - 6 = 0$$

If $x = \alpha$ is a root of the original equation and we want an equation with the root $w = 2\alpha + 3$, then we can rearrange to get $\alpha = \frac{w-3}{2}$ and sub $\frac{w-3}{2}$ into the original equation.

$$3\left(\frac{w-3}{2}\right)^2 - 4\left(\frac{w-3}{2}\right) - 6 = 0$$
$$\frac{3}{4}w^2 - \frac{9}{2}w + \frac{27}{4} + 6 - 2w - 6 = 0$$
$$\frac{3}{4}w^2 - \frac{13}{2}w + \frac{27}{4} = 0$$
$$3w^2 - 26w + 27 = 0$$

1 d)

$$3x^2 - 4x - 6 = 0$$

If $x = \alpha$ is a root of the original equation and we want an equation with the root $w = \frac{3}{\alpha}$, then we can rearrange to get $\alpha = \frac{3}{w}$ and sub $\frac{3}{w}$ into the original equation.

$$3\left(\frac{3}{w}\right)^2 - 4\left(\frac{3}{w}\right) - 6 = 0$$

$$\frac{27}{w^2} - \frac{12}{w} - 6 = 0$$
$$27 - 12w - 6w^2 = 0$$
$$2w^2 + 4w - 9 = 0$$

2 a)

$$3z^3 - 3z^2 + 8z + 5 = 0$$

If $z = \alpha$ is a root of the original equation and we want an equation with the root $w = 2\alpha$, then we can rearrange to get $\alpha = \frac{w}{2}$ and sub $\frac{w}{2}$ into the original equation.

$$3\left(\frac{w}{2}\right)^3 - 3\left(\frac{w}{2}\right)^2 + 8\left(\frac{w}{2}\right) + 5 = 0$$
$$\frac{3}{8}w^3 - \frac{3}{4}w^2 + 4w + 5 = 0$$
$$3w^3 - 6w^2 + 32w + 40 = 0$$

2 b)

$$3z^3 - 3z^2 + 8z + 5 = 0$$

If $z = \alpha$ is a root of the original equation and we want an equation with the root $w = 3 - \alpha$, then we can rearrange to get $\alpha = 3 - w$ and sub $3 - w$ into the original equation.

$$3(3-w)^3 - 3(3-w)^2 + 8(3-w) + 5 = 0$$
$$-3w^3 + 27w^2 - 81w + 81 - 3w^2 + 18w - 27 + 24 - 8w + 5 = 0$$
$$3w^3 - 24w^2 + 71w - 83 = 0$$

2 c)

$$3z^3 - 3z^2 + 8z + 5 = 0$$

If $z = \alpha$ is a root of the original equation and we want an equation with the root $w = 2\alpha - 1$, then we can rearrange to get $\alpha = \frac{w+1}{2}$ and sub $\frac{w+1}{2}$ into the original equation.

$$3\left(\frac{w+1}{2}\right)^3 - 3\left(\frac{w+1}{2}\right)^2 + 8\left(\frac{w+1}{2}\right) + 5 = 0$$
$$\frac{3}{8}(w+1)^3 - \frac{3}{4}(w+1)^2 + 4(w+1) + 5 = 0$$
$$\frac{3}{8}w^3 + \frac{9}{8}w^2 + \frac{9}{8}w + \frac{3}{8} - \frac{3}{4}w^2 - \frac{3}{2}w - \frac{3}{4} + 4w + 4 + 5 = 0$$
$$\frac{3}{8}w^3 + \frac{3}{8}w^2 + \frac{29}{8}w + \frac{69}{8} = 0$$
$$3w^3 + 3w^2 + 29w + 69 = 0$$

2 d)

$$3z^3 - 3z^2 + 8z + 5 = 0$$

If $z = \alpha$ is a root of the original equation and we want an equation with the root $w = \frac{\alpha}{3} + 1$, then we can rearrange to get $\alpha = 3w - 3$ and sub $3w - 3$ into the original equation.

$$3(3w-3)^3 - 3(3w-3)^2 + 8(3w-3) + 5 = 0$$
$$81w^3 - 243w^2 + 243w - 81 - 27w^2 + 54w - 27 + 24w - 24 + 5 = 0$$
$$81w^3 - 270w^2 + 321w - 127 = 0$$

2 e)

$$3z^3 - 3z^2 + 8z + 5 = 0$$

If $z = \alpha$ is a root of the original equation and we want an equation with the root $w = \frac{1}{2\alpha}$, then we can rearrange to get $\alpha = \frac{1}{2w}$ and sub $\frac{1}{2w}$ into the original equation.

$$3\left(\frac{1}{2w}\right)^3 - 3\left(\frac{1}{2w}\right)^2 + 8\left(\frac{1}{2w}\right) + 5 = 0$$
$$\frac{3}{8w^3} - \frac{3}{4w^2} + \frac{4}{w} + 5 = 0$$
$$\frac{3}{8} - \frac{3}{4}w + 4w^2 + 5w^3 = 0$$
$$40w^3 + 32w^2 - 6w + 3 = 0$$

3 a)

$$w = 2\alpha \Rightarrow \alpha = \frac{w}{2}$$

$$2\left(\frac{w}{2}\right)^4 - 2\left(\frac{w}{2}\right)^3 + \left(\frac{w}{2}\right)^2 + 4\left(\frac{w}{2}\right) + 7 = 0$$
$$\frac{w^4}{8} - \frac{w^3}{4} + \frac{w^2}{4} + 2w + 7 = 0$$
$$w^4 - 2w^3 + 2w^2 + 16w + 56 = 0$$

3 b)

$$w = 4 - \alpha \Rightarrow \alpha = 4 - w$$

$$2(4-w)^4 - 2(4-w)^3 + (4-w)^2 + 4(4-w) + 7 = 0$$
$$2w^4 - 32w^3 + 192w^2 - 512w + 512 + 2w^3 - 24w^2$$
$$+ 96w - 128 + w^2 - 8w + 16 + 16 - 4w + 7 = 0$$
$$2w^4 - 30w^3 + 169w^2 - 428w + 423 = 0$$

3 c)

$$w = 2\alpha + 3 \Rightarrow \alpha = \frac{w-3}{2}$$

$$2\left(\frac{w-3}{2}\right)^4 - 2\left(\frac{w-3}{2}\right)^3 + \left(\frac{w-3}{2}\right)^2 + 4\left(\frac{w-3}{2}\right) + 7 = 0$$
$$\frac{1}{8}(w-3)^4 - \frac{1}{4}(w-3)^3 + \frac{1}{4}(w-3)^2 + 2(w-3) + 7 = 0$$
$$\frac{1}{8}w^4 - \frac{3}{2}w^3 + \frac{27}{4}w^2 - \frac{27}{2}w + \frac{81}{8} - \frac{1}{4}w^3 + \frac{9}{4}w^2$$
$$- \frac{27}{4}w + \frac{27}{4} + \frac{1}{4}w^2 - \frac{3}{2}w + \frac{9}{4} + 2w - 6 + 7 = 0$$
$$\frac{1}{8}w^4 - \frac{7}{4}w^3 + \frac{37}{4}w^2 - \frac{79}{4}w + \frac{161}{8} = 0$$
$$w^4 - 14w^3 + 74w^2 - 158w + 161 = 0$$

3 d)

$$w = \frac{1}{\alpha} \Rightarrow \alpha = \frac{1}{w}$$

$$2\left(\frac{1}{w}\right)^4 - 2\left(\frac{1}{w}\right)^3 + \left(\frac{1}{w}\right)^2 + 4\left(\frac{1}{w}\right) + 7 = 0$$
$$\frac{2}{w^4} - \frac{2}{w^3} + \frac{1}{w^2} + \frac{4}{w} + 7 = 0$$
$$2 - 2w + w^2 + 4w + 7w^4 = 0$$
$$7w^4 + 4w^3 + w^2 - 2w + 2 = 0$$

4 a)

$$2z^2 + 5z + k = 0$$

If $z = \alpha$ is a root of the original equation and we want an equation with the root $w = \frac{\alpha}{2} - 1$, then we can rearrange to get $\alpha = 2w + 2$ and sub $2w + 2$ into the original equation.

$$2(2w+2)^2 + 5(2w+2) + k = 0$$
$$2(4w^2 + 8w + 4) + 10w + 10 + k = 0$$
$$8w^2 + 26w + 18 + k = 0$$

4 b)

$$8w^2 + 26w + \frac{185}{8} = 0$$

which gives $w = -\frac{13}{8} \pm \frac{1}{2}i$

$$\alpha = 2\left(-\frac{13}{8} + \frac{1}{2}i\right) + 2 = -\frac{5}{4} + i$$

and $\alpha = 2\left(-\frac{13}{8} - \frac{1}{2}i\right) + 2 = -\frac{5}{4} - i$

5 a)

$$x^3 + 2x^2 - 10x + k = 0$$

If $x = \alpha$ is a root of the original equation and we want an equation with the root $w = \alpha - 2$, then we can rearrange to get $\alpha = w + 2$ and sub $w + 2$ into the original equation.

$$(w+2)^3 + 2(w+2)^2 - 10(w+2) + k = 0$$
$$w^3 + 6w^2 + 12w + 8 + 2w^2 + 8w + 8 - 10w - 20 + k = 0$$
$$w^3 + 8w^2 + 10w - 4 + k = 0$$

5 b)

Since the curve $y = g(x)$ is a translation of the curve $y = f(x)$, the stationary points have translated.

So the stationary points occur when $x = a - 2$ and $x = b - 2$

6 a)

$$2z^3 - z^2 + 3z + k = 0$$

If $z = \alpha$ is a root of the original equation and we want an equation with the root $w = \frac{\alpha}{2} + 1$, then we can rearrange to get $\alpha = 2w - 2$ and sub $2w - 2$ into the original equation.

$$2(2w-2)^3 - (2w-2)^2 + 3(2w-2) + k = 0$$
$$16w^3 - 48w^2 + 48w - 16 - 4w^2 + 8w - 4 + 6w - 6 + k = 0$$
$$16w^3 - 52w^2 + 62w - 26 + k = 0$$

6 b)

$$\alpha = 2 \times \frac{3}{4} - 2 = -\frac{1}{2}$$

7 a)

$$\alpha\beta = -7$$

7 b)

$$\left(\frac{y}{\alpha\beta}\right)^2 + 5\left(\frac{y}{\alpha\beta}\right) - 7 = 0$$
$$\left(\frac{y}{-7}\right)^2 + 5\left(\frac{y}{-7}\right) - 7 = 0$$
$$\frac{1}{49}y^2 - \frac{5}{7}y - 7 = 0$$
$$y^2 - 35y - 343 = 0$$

8)

$$x^3 + kx^2 + (2k-5)x + 10 = 0$$

If $x = \alpha$ is a root of the original equation and we want an equation with the root $w = \alpha + 3$, then we can rearrange to get $\alpha = w - 3$ and sub $w - 3$ into the original equation.

$$\begin{aligned}(w-3)^3 + k(w-3)^2 + (2k-5)(w-3) + 10 &= 0\\ w^3 - 9w^2 + 27w - 27 + kw^2 - 6kw + 9k&\\ + 2kw - 5w - 6k + 15 + 10 &= 0\\ w^3 + (k-9)w^2 + (22-4k)w + (3k-2) &= 0\end{aligned}$$